



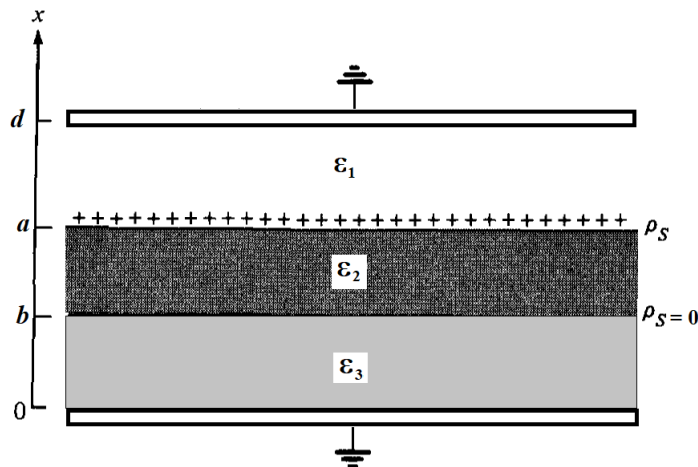
Midterm II Exam

EE241: Electromagnetic Theory
Instructor: Dr. Shafayat Abrar
Marks: 50 [20%]

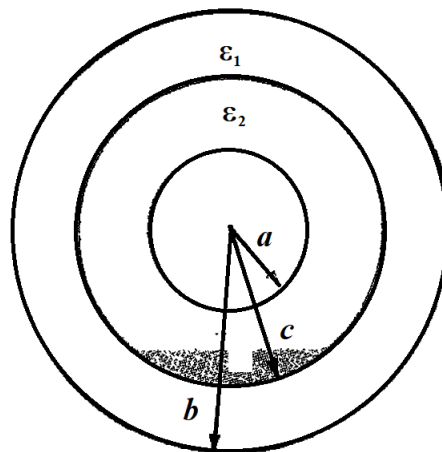
Date: April 29, 2019
Term: Spring 19
Duration: 115 minutes

[i] Attempt all questions. This is an open book exam. No talking; no cheating. Provide all necessary steps. No query will get attention.

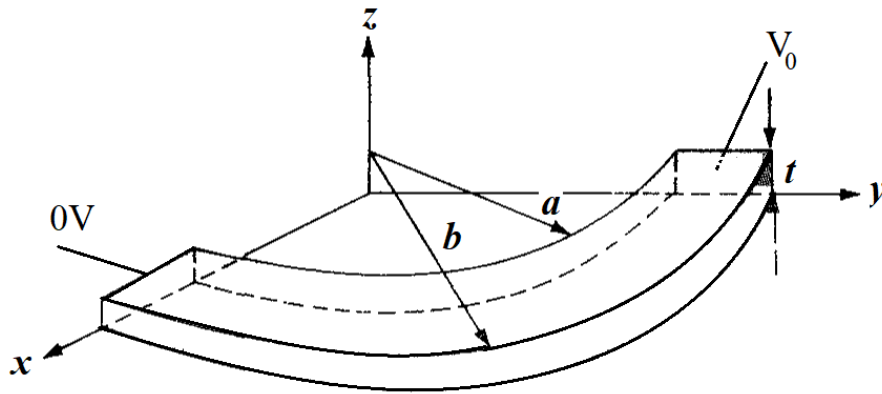
- [14 points]** Refer to figure below, where the upper and lower electrodes are both grounded. The surface charge between dielectric material 1 and 2 is $\rho_s \neq 0$; while there is no surface charge between dielectric material 2 and 3. Find electric field in all three inner regions.



- [12 points]** The cross section of a spherical capacitor is shown below, determine its capacitance given $a = 2 \text{ mm}$, $b = 6 \text{ mm}$, $c = 4 \text{ mm}$, $\epsilon_{r1} = 3$ and $\epsilon_{r2} = 4$.



3. [12 points] A metal bar of conductivity σ is bent to form a flat 90° sector of inner radius a , outer radius b , and thickness t as shown below. Find the resistance of the bar between the vertical rectangular surfaces at $\phi = 0$ and $\phi = \pi/2$.



4. [12 points] A metal sphere of radius b has a uniform surface charge distribution. This metallic sphere is surrounded by a dielectric medium whose permittivity varies as $\epsilon = \epsilon_0(1 + a/r)$. Show that the potential in dielectric region is

$$V = \frac{Q}{4\pi\epsilon_0 a} \ln\left(1 + \frac{a}{r}\right)$$

where Q is the total charge on the sphere.

Q1 Solution

Pre-Final Exam Solution
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For each region, we solve Laplace's equation to obtain

$$V_1(x) = A_1 x + B_1 \quad ; \quad a \leq x \leq d$$

$$V_2(x) = A_2 x + B_2 \quad ; \quad b \leq x \leq a$$

$$V_3(x) = A_3 x + B_3 \quad ; \quad 0 \leq x \leq b$$

Electric fields are obtained as:

$$\bar{E}_1 = -\frac{\partial V_1}{\partial x} \hat{a}_x = -A_1 \hat{a}_x$$

$$\bar{E}_2 = -\frac{\partial V_2}{\partial x} \hat{a}_x = -A_2 \hat{a}_x$$

$$\bar{E}_3 = -\frac{\partial V_3}{\partial x} \hat{a}_x = -A_3 \hat{a}_x$$

Boundary conditions yield:

$$\text{@ } x = d \quad V_1 = 0 \quad \text{_____} (1)$$

$$\text{@ } x = a \quad V_1 = V_2 \quad \text{_____} (2)$$

$$\text{@ } x = b \quad V_2 = V_3 \quad \text{_____} (3)$$

$$\text{@ } x = 0 \quad V_3 = 0 \quad \text{_____} (4)$$

$$\text{@ } x = a \quad \epsilon_1 E_1 - \epsilon_2 E_2 = \rho_s \quad \text{_____} (5)$$

$$\text{@ } x = b \quad \epsilon_2 E_2 - \epsilon_3 E_3 = 0 \quad \text{_____} (6)$$

$$E_q(1) \Rightarrow A_1 d + B_1 = 0$$

$$E_q(2) \Rightarrow A_1 a + B_1 = A_2 a + B_2$$

$$E_q(3) \Rightarrow A_2 b + B_2 = A_3 b + B_3$$

$$E_q(4) \Rightarrow A_3(0) + B_3 = 0 \Rightarrow B_3 = 0$$

$$E_q(5) \Rightarrow -\epsilon_1 A_1 + \epsilon_2 A_2 = f_s$$

$$E_q(6) \Rightarrow -\epsilon_2 A_2 + \epsilon_3 A_3 = 0$$

Solving the above, we obtain:

$$A_1 = -f_s \frac{P_2}{P_1}$$

$$A_2 = -\frac{a\epsilon_3 f_s - d\epsilon_3 f_s}{P_1}$$

$$A_3 = -\frac{\epsilon_2 f_s (a-d)}{P_1}$$

$$B_1 = \frac{d f_s P_2}{P_1}$$

$$B_2 = -\frac{b f_s (a\epsilon_2 - a\epsilon_3 - d\epsilon_2 + d\epsilon_3)}{P_1}$$

where

$$P_1 = a\epsilon_1\epsilon_3 - a\epsilon_2\epsilon_3 + b\epsilon_1\epsilon_2 - b\epsilon_1\epsilon_3 + d\epsilon_2\epsilon_3$$

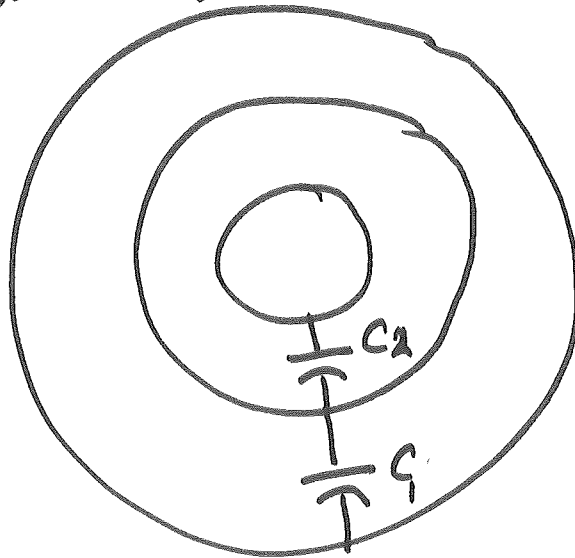
$$P_2 = a\epsilon_3 + b\epsilon_2 - b\epsilon_3$$

Solution

Q2. The spherical capacitor has following capacitance (proof was not required)

$$C = \frac{4\epsilon_0\epsilon_0}{\frac{1}{a} - \frac{1}{b}}, \quad b > a$$

The given configuration has two capacitance connected in series.



$$C_1 = \frac{4\epsilon_0\epsilon_{r1}\pi}{\frac{1}{c} - \frac{1}{b}}$$

$$C_2 = \frac{4\epsilon_0\epsilon_{r2}\pi}{\frac{1}{a} - \frac{1}{c}}$$

$$C_{eq} = \frac{C_1 \times C_2}{C_1 + C_2}$$

[F]

[You can substitute the values and obtain C_{eq}].

Q3. Solution
 $\nabla_{\phi}^2 V = 0$ gives $V = A\phi + B$

$$V(\phi=0) = 0$$

$$V(\phi=\pi/2) = V_0$$

$$\Rightarrow B=0 \Rightarrow V = \left(\frac{V_0}{\pi/2}\right)\phi, \quad 0 \leq \phi \leq \frac{\pi}{2}$$

$$\vec{E} = -\nabla V = \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{\phi} = \frac{2V_0}{\pi\rho} \hat{\phi}$$

$$I = \int \vec{J} \cdot d\vec{S} = \int \sigma \vec{E} \cdot d\vec{S}$$

$$d\vec{S} = dz d\rho \hat{\phi}$$

$$I = \frac{2\sigma V_0}{\pi} \int_{\rho=a}^b \int_{z=0}^t \frac{1}{\rho} dz d\rho$$

$$= \frac{2\sigma V_0}{\pi} \ln \rho \Big|_a^b \cdot z \Big|_0^t = \frac{2\sigma V_0 t}{\pi} \ln\left(\frac{b}{a}\right)$$

$$R = \frac{V_0}{I} = \frac{\pi}{2\sigma t \ln(b/a)}$$

$[\Omega]$

Q4. Solution

$$d\vec{E} = \frac{dq}{4\pi\epsilon r^2} \hat{a}_r$$

where $\epsilon = \epsilon_0(1 + a/r)$

and $dq = \rho_s ds$

$$ds = b^2 \sin\theta d\theta d\phi$$

$$\vec{E} = \int d\vec{E} = \frac{\int dq}{4\pi\epsilon r^2} \hat{a}_r$$

$$\int dq = \rho_s \iiint b^2 \sin\theta d\theta d\phi = \rho_s b^2 \int_{\phi=0}^{2\pi} \int_{\theta=0}^{\pi} \sin\theta d\theta d\phi$$

$$= 4\pi b^2 \rho_s$$

$\therefore Q = \text{Total Charge}$

$$\Rightarrow \vec{E} = \frac{Q \hat{a}_r}{4\pi\epsilon r^2} = \frac{Q \hat{a}_r}{\epsilon_0 4\pi (1 + \frac{a}{r}) r^2} = \frac{Q \hat{a}_r}{\epsilon_0 4\pi r(r+a)}$$

$$\vec{E} \cdot d\vec{l} = \frac{Q dr}{\epsilon_0 4\pi r(r+a)} = -\frac{Q}{\epsilon_0 4\pi a} \left(\frac{1}{r+a} - \frac{1}{r} \right)$$

$$V = - \int_{\infty}^r \vec{E} \cdot d\vec{l} = \frac{Q}{\epsilon_0 4\pi a} \int_{\infty}^r \left(\frac{1}{r'+a} - \frac{1}{r'} \right) dr'$$

$$= \frac{Q}{\epsilon_0 4\pi a} \left(\ln(r'+a) - \ln(r') \right) \Big|_{\infty}^r \Rightarrow \boxed{V = \frac{Q}{\epsilon_0 4\pi a} \ln\left(1 + \frac{a}{r}\right)}$$

[volt] for $r \geq b$